

Retail Profit Optimization & Sales Forecasting

Executive Overview

Objective

Analyze retail sales, profit, inventory, promotions, and customer behavior to identify profit leakage and improve business performance using Python, SQL, Power BI, and Tableau.

Business Problems Identified

- High discounts reducing net profit
 - Frequent product returns impacting margins
 - Stockouts causing revenue loss
 - Uneven store performance
 - Some promotions increasing sales but lowering profitability
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Technologies Used

- Python → Data Cleaning & Forecasting
 - SQL → KPI & Business Analytics
 - Power BI → Interactive Dashboard
 - Tableau → Executive Storytelling Dashboard
 - Excel → Raw Data Source
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Key KPIs Analyzed

- Total Sales
 - Net Profit
 - Profit Margin %
 - Return Rate
 - Discount Leakage
 - Stockout Loss
 - Forecast Accuracy
 - Customer Segments
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Major Findings

- Profit declined after heavy discount campaigns
- Some high-sales stores had low margins
- Returns significantly affected profitability
- Inventory delays increased stockout risk
- Forecasting identified future sales trends

Business Insights

Store Performance

Top-performing stores generated high revenue but suffered reduced profit margins due to excessive discounts and returns.

Promotion Analysis

Certain promotions improved sales volume but negatively impacted net profitability.

Inventory Insights

Vendor delays and stock shortages created missed sales opportunities.

Customer Analysis

Repeat customers contributed significantly to overall revenue.

Forecasting Insights

Sales forecasting predicted seasonal demand fluctuations and future revenue trends, helping improve inventory planning and decision-making.

Recommendations

- Reduce unnecessary discount campaigns
 - Improve return management policies
 - Optimize inventory replenishment
 - Focus on profitable promotions
 - Improve vendor delivery efficiency
 - Target loyal customer segments
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Final Outcome

The project successfully identified major profit leakage areas and provided data-driven strategies to improve profitability, operational efficiency, and future sales planning.